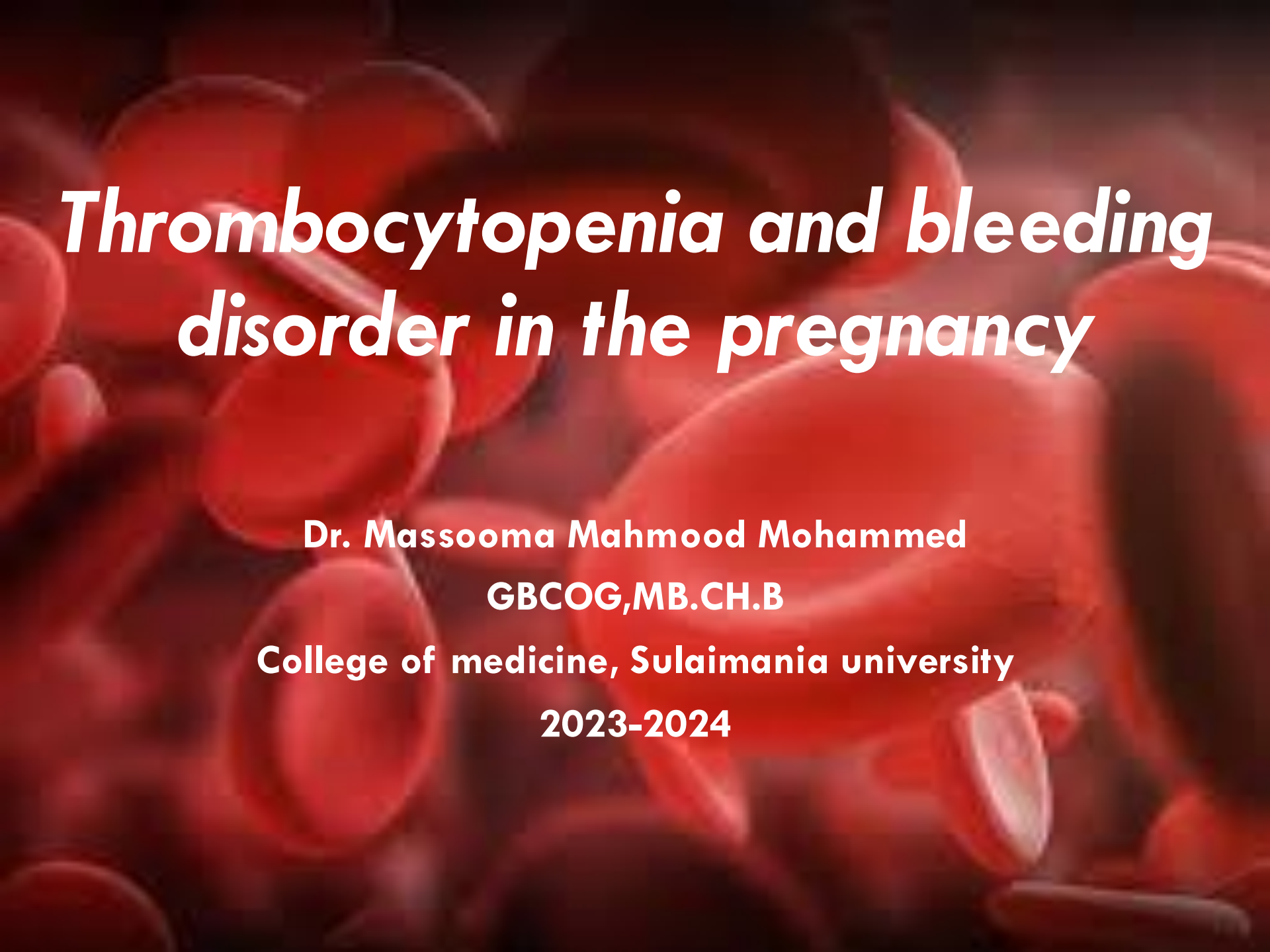


A microscopic view of numerous red blood cells, which are biconcave discs, filling the frame. They are rendered in various shades of red and pink, with some appearing more prominent than others due to focus and lighting. The background is a soft, out-of-focus red.

Thrombocytopenia and bleeding disorder in the pregnancy

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Normal haemostasis & the vascular tree

The normal function of coagulation & fibrinolytic system is to maintain an intact but patent vascular tree.

Three main component plays a part in normal haemostasis:

- 1.Vascular constriction.**
- 2.Platelet plug .**
- 3.Fibrin generation**

The fibrinolytic system is complementary to these activities & is responsible for the removal of fibrin & the restoration of vascular patency.

Failure of normal fibrin formation

1. Insufficient fibrinogen
2. Deficiencies in one or more of clotting factors
3. Failure of normal fibrin stabilization.
4. FDPs (fibrin degradation products).

Coagulation Inhibitors:

Anti- thrombin III (ATIII)

- Alpha 2 globulin inhibits Thrombin & factors Xa, XIa, XIIa, XIa, IXa.
- Protein C (endothelial cell) .
- Protein S (endothelial cell & platelets) .

Coagulation & fibrinolytic system during pregnancy

- Placental separation during the 3rd stage of labour represents a major haemostatic challenge to the mother
- Physiological adaptations occur during pregnancy to help the mother meet this haemostatic challenge .
- Together the change in coagulation & fibrinolysis in pregnancy represents a **hypercoagulable state** .

Fibrinolytic system during pregnancy

- - Plasma plasminogen levels rise in tandem with the rise of fibrinogen.**
 - By contrast ,the euglobulin lysis time ,which measures plasminogen activator activity , is markedly prolonged .**
 - Anti plasmins also rise so that the capacity to generate plasmin may be reduced in pregnancy .**

Coagulation & fibrinolysis during puerperium

- Following delivery , major changes occur in the coagulation & fibrinolytic system .**
- Rise in plasminogen activator activity which return to non pregnant range within 30 min of delivery .**
- Fibrinogen level & platelets count rise during early puerperium.**
- Anti- thrombin activity increase .**
- Following the initial phase o f increased clotting factors in the puerperium ,the coagulation & fibrinolytic system gradually revert to normal within 6 weeks after delivery.**

Disorders of hemostasis:

Disorders of the platelet and vessel wall

Immune thrombocytopenic purpura(ITP)

Thrombotic thrombocytopenic purpura(TTP)

Hemolytic-uremic syndrome(HUS)

Paroxysmal nocturnal hemoglobinuria

Platelet Dysfunction

Disorders of coagulation and thrombosis and bleeding disorders

-Disseminated intravascular coagulation

-Factor deficiencies

Hemophilia A (Factor VIII deficiency)

Hemophilia B (Factor IX deficiency, "Christmas disease")

Hemophilia C (Factor XI deficiency, mild bleeding tendency)

Von Willebrand disease(the most common bleeding disorder)

Disorders predisposing to thrombosis

-Heparin-induced thrombocytopenia("white clot syndrome)

-Antiphospholipid syndrome

- **Lupus anticoagulant**
- **Anticardiolipin antibody**

-Factor V Leiden and Activated Protein C Resistance

-Prothrombin mutation

-Protein C deficiency

-Protein S deficiency

-Antithrombin deficiency

-Abnormally raised levels of Factor VIII and Factor XI

Bleeding disorders:

Bleeding disorders is either inherited or acquired:

Inherited bleeding disorders:

1.VON Willebrand's disease (VWD).

von willebrand factor is a plasma protein that has two main functions:

- Stabilization of factor VIII
- Adherence of platelet to injured vessel walls
- Generally, this is inherited as an autosomal dominant condition, although there are recessive variants.
- It is the most common inherited bleeding disorder.

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- Levels of von Willebrand factor can be normal in pregnancy because of the increased production in the liver, but they return to pre-pregnancy values by three days post partum and that is why it is actually delayed post partum hemorrhage that is more of an issue.**
 - In women, menorrhagia and delayed post partum hemorrhage are common presentations**
 - Other clinical manifestations include bleeding, epistaxis, gingival bleeding**

Hemophilia

The next inherited bleeding disorder which are uncommon in females are

- hemophilia A: which** is due to factor VIII deficiency and it is an X-linked recessive and females are usually carrier for the disease.
- Hemophilia B** is also known as Christmas disease; it is factor IX deficiency. Again, it is an X-linked recessive and it is much less common than hemophilia A.
- Factor XI deficiency.**

Thrombocytopenia

Thrombocytopenia is a reduction in platelet number below 150 000/mcl

Causes:

- 1- incidental thrombocytopenia of pregnancy
- 2- increased consumption
- 3- autoimmune thrombocytopenia (ITP).
- 4- SLE/APS
- 5-Activated clotting mechanism
 - Pre-eclampsia
 - HELP syndrome
 - DIC
- 6-Thrombotic thrombocytopenic purpura
- 7-Decreased platelet production (marrow suppression)
 - sepsis
 - HIV
- 8- Malignant marrow infiltration

Idiopathic Thrombocytopenic Purpura.

Idiopathic thrombocytopenic purpura (ITP) is a common autoimmune disorder in which patients form antiplatelet autoantibodies against platelet-specific antigen

- Often the patients are asymptomatic and pregnancy does not always exacerbate the disease.**
- If platelet count is more than 500000/microliter no treatment is necessary.**
- Major bleeding is rarely seen unless the platelet count is <10000/mcl**
- Maternal antibodies may cross the placenta and affect the fetus , causing neonatal thrombocytopenia.**
- four to ten percent of neonates are at risk of having severe thrombocytopenia at birth or during the 1st week of life.**
- Traumatic vaginal delivery must be avoided.**
- Platelet count and pediatric assessment is indicated and the infant's platelet count followed carefully over the next week.**

Management of ITP

- Maternal indications for treatment of thrombocytopenia should not differ from those for non pregnant individuals.**
- Therapy is not initiated unless platelets are $< 50\ 000/\mu\text{L}$ or potential hemorrhages are present**
- Corticosteroid 1mg/kg per day of prednisolone are given initially, maintained for 2-3 weeks then tapered slowly**
- Intravenous immunoglobulin can be given for corticosteroid failure in Rh –positive women.**
- Splenectomy is the last resort for patients who fail to respond to corticosteroid or immunoglobulin treatment.**
- Platelet transfusion are not recommended except in life threatening situation.**

The acquired disorders that lead to bleeding.

These include:

DIC, vitamin K deficiency, liver disease, uremia and after massive transfusion.

DIC(Disseminated intravascular coagulation)

- The name of this disorder arises from the fact that malfunction of clotting factors cause platelets to clot in small blood vessels throughout the body.
- This action leads to a lack of clotting factors and platelets at a site of injury that requires clotting.
- Patients with disseminated intravascular coagulation (DIC) will bleed abnormally even though there is no history of coagulation abnormality.
- Symptoms may include minute spots of hemorrhage on the skin, and purple patches.
- A patient may bleed from surgery or intravenous injection (IV) sites. Related symptoms include vomiting, seizures, coma, shortness of breath, shock, severe pain in the back, muscles, abdomen, or chest.

-DIC is not a hereditary disorder or a common one.

-It is most commonly caused by:

1-complications during pregnancy or delivery:

- Abruptio placenta.**
- amniotic fluid embolus.**
- severe preeclampsia.**
- retained dead fetus.**
- sepsis.**
- second-trimester abortion.**

2-overwhelming infections,

3-acute leukemia, metastasis cancer.

4-extensive burns and trauma.

5-even snakebites.

-Thrombi in the microcirculation activates the fibrinolytic process and leads to the release of fibrin degradation products, which inhibit normal coagulation. The consumption of platelets and coagulation factors, as well as the above described inhibition of normal coagulation, leads to both hemorrhagic and thrombotic consequences.

laboratory investigations:

- Fibrin degradation products high.
- Serum fibrinogen- low
- Prothrombin time (PT) -prolonged
- Partial thromboplastin time(PTT) - prolonged
- Platelet count: low

Treatment of disseminated intravascular coagulopathy

- Urgent haematological consultation
 - Monitoring in intensive care unite
- Check platelet count
- Give cryoprecipitate & fresh frozen plasma
- Transfuse with fresh blood if available
- Treatment of the underlying cause.

Thrombophilia :

Is defined as a predisposition to thrombosis, secondary to any persistent or identifiable hypercoagulable state.

It can be inherited or acquired:

-It should be considered in :

- a young patient who experiences atraumatic thrombosis.**
- patients who have a family history of thrombosis.**
- cases of recurrent thrombosis, especially when someone is already anti-coagulated**
- when thrombosis occurs at an unusual site.**
- It should also be considered, for our purposes, in patients who have recurrent pregnancy loss, unexplained IUFD's and early severe IUGR.**

Causes of thrombophilia

Inherited causes

1. Anti thrombin III deficiency:

- Anti thrombin III is a naturally occurring anticoagulant.**
- It inactivate thrombin and factors IXa, Xa, XIa and XIIa.**
- This is an autosomal-dominant condition.**

- The clinical manifestation is thrombosis.**

- It may be an acquired deficiency in patients who have DIC, nephrotic syndrome, liver disease, pre-eclampsia, during oral contraceptive use and during heparin therapy.**

2. Protein C deficiency

-is also autosomal dominant

-This is the next thrombophilia.

3-Protein S deficiency is also autosomal dominant.

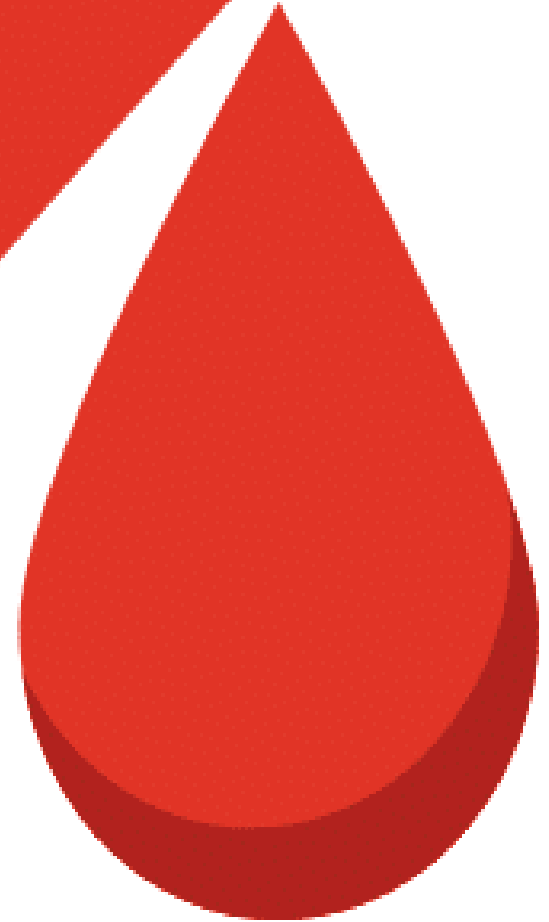
4-Factor V Leiden mutation.

5.Prothrombin gene mutation.

Acquired causes of thrombophilia is Antiphospholipid syndrome

MARCH IS BLEEDING DISORDER AWARENESS MONTH

World
Hemophilia Day
is April 17





***THANK
YOU***